The H-LAYER residual inventory after the STEP-5B conditional closure: six named items between the amended-class theorem and unconditional

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

GATES.md records H-LAYER's residual as "exactly STEP-5B." With STEP-5B flipped to CLOSED-CONDITIONAL, the honest question for the T7 trajectory is: what EXACTLY remains between the amended-class conditional theorem (unit U1) and an unconditional whole-Reading-H statement? This note answers by decomposition into six named items. No new theorem is proved; every item is assigned its honest current tier from material already on disk.

2. The six named residuals

Item	Content and current protection	Honest status / discharge
RES-1 (H-diag closure)	Math427's isotropy-infimum theorem is conditional on the DIAGONAL-Gaussian restriction; off-diagonal (Bloch) covariance refinements were executed for the enumerated five (Math428) and are bounded class-wide ON THE AMENDED CLASS by the B5 Lemma-B second-order log-det envelope.	Covered conditionally at second order (SC-SCOPE); the beyond-second-order log-det remainder merges with RES-5. Discharge: a log-det envelope with explicit third-order remainder bound.
RES-2 (sigma inhomogeneity)	$\sigma(x)$-inhomogeneous dressing was executed for the enumerated five (Math429) and its channel is closed in the B5 G2 bookkeeping (Lemma I: σ-channel completeness, exact to 10^{-12}) on the amended class.	Covered conditionally at second cumulant. Discharge: folds into the same SC-SCOPE lift as RES-1.
RES-3 (unrestricted class)	The H-ADM-COH amendment quotients out sub-resolution structure; energy-faithfulness (AddC, $c_{\text{ind}} = 30.1$; AddE depth-free $c_{\text{total}} \leq 20$) makes the quotient canonical – every unrestricted pattern has an amended representative within $c_{\text{ind}} I^2$ of its energy, which is margin/ $\times$ 33 or smaller.	The genuine combinatorial frontier: the UNCONDITIONAL class-wide energy bound is DR-2 (sphere structure dichotomy), assessed research-grade (adjacent to the open circle-incidence conjecture). Off the critical path by verdict #14; the publication-strength alternative.
RES-4 (intensity interval)	The margin chain is certified at the three anchor intensities $\{4\times10^{-4}, 10^{-3}, 2\times10^{-3}\}$; in between, every ingredient varies monotonically or mildly ($\hat{r}$ by $< 10\%$; budgets monotone in I).	Three-point certification only. Discharge: an interval-arithmetic sweep of the margin chain over $I \in [4\times10^{-4}, 2\times10^{-3}]$ (mechanical; registered polish; candidate follow-up script).
RES-5 (matched-order to exact)	All comparisons live at matched second-cumulant bookkeeping: the same functional, same order, for every state. Converting "$F^{(2)}[\mathcal{P}] > F^{(2)}[\mathcal{R}_H]$" into a statement about EXACT free energies requires either a controlled error bound (GAP-2 / ESTIMATOR-UPGRADE machinery) or acceptance of the matched-order functional as the selection definition (the current pinned scope). The third-cumulant lift assessment (unit U4) quantifies the first correction.	The deepest analytic frontier; aligns 1:1 with GAP-2. Discharge: third-cumulant bound at the anchor (U4 gives the size assessment) + an order-remainder envelope, or the GAP-2 controlled-error programme.
RES-6 (sea-sector completeness)	H-ADM-COH reclassifies sub-resolution mass into the Gaussian sea; the layer's isotropic dressing is the sea accounting. The variational completeness of that accounting (no sub-resolution configuration escapes both the reading ledger and the sea dressing) is exactly what AddC's lemma certifies at second cumulant ($\lvert F[P'] - F[P]\rvert \leq c_{\text{ind}} I^2$).	Covered conditionally; the unconditional form merges with RES-3 (the quotient is canonical iff the energy-faithfulness holds class-wide, which the AddC/AddE chain gives on the amended scope).

3. Structural summary

- **Conditionally covered now** (by the existing certified chain, on the amended class at second cumulant, three anchors): RES-1, RES-2, RES-6, and the amended-class core that used to be "exactly STEP-5B."
- **Mechanical polish**: RES-4 (interval sweep).
- **Genuine open frontiers**: RES-3 (unrestricted class = DR-2, research-grade combinatorics) and RES-5 (matched-order to exact = GAP-2 analysis). These two are INDEPENDENT axes: closing either one alone strengthens the conditional theorem; closing both (plus RES-4) is what "unconditional whole-Reading-H at the operating point" means.
- **Consequence for the U1 hypothesis set**: H-LAYER$^\flat$ decomposes as {layer framing = RES-5 axis} $\oplus$ {class restriction = RES-3 axis} $\oplus$ {order restriction = SC-SCOPE, quantified by U4}; H-ADM-COH is the RES-3 axis's conditional discharge. The four-hypothesis set of U1 is therefore irredundant: no listed hypothesis is implied by the others.

4. Numerical verification

No new machine numbers: every figure quoted (margins, $c_{\text{ind}} = 30.1$, $c_{\text{total}} \leq 20$, $\times 33$ floor, $< 10\%$ $\hat{r}$ variation, 10^{-12} σ-channel exactness) is already certified in the existing artefact `claims/B5-BEYOND-LAYER-BOUND/runs/260605-gershgorin-reduction/result.json` (script v1.14.0, 192/192) and the migrated Math427–Math429 records.

Quantitative sanity check (mandatory): consistency cross-check of the decomposition — the six items partition the logical distance with no overlap double-counted: RES-1/RES-2 are order-limited (their residual IS the SC-SCOPE lift), RES-6 reduces to RES-3, leaving exactly three independent axes {RES-3, RES-4, RES-5}; this matches the U1 exclusion list one-to-one (unrestricted class, off-anchor intensities, third-cumulant order) plus the two standing gates (G3PB-III multishell and ROBUSTNESS-MU2 are OPERATING-POINT axes, listed in U1 separately and deliberately NOT re-counted here). PASS: no residual named in U1 is missing from this inventory's union with the U1 operating-point list, and vice versa.

5. Devil's-advocate (three objections; CLAUDE.md 6.3)

(α) *"RES-1's claim that Lemma B covers off-diagonal Bloch class-wide is an overreach: Lemma B is a second-order envelope, and Math427's H-diag is about the TRIAL CLASS, not the correction order."* VALID-with-mitigation: agreed that the two cuts (trial-class vs correction-order) are conceptually distinct; the inventory's claim is precisely scoped — the off-diagonal ENERGY CORRECTION is what enters the selection inequality, and that correction is what Lemma B bounds at second order on the amended class. The trial-class infimum question beyond diagonal-Gaussian (could a full-covariance Gaussian trial LOWER a competitor's layer value below the Prop-A floor?) is the same mathematical object viewed from the variational side, and its beyond-second-order content is exactly the RES-5 merger stated in the table. The wording in RES-1 carries this merger explicitly.

(β) *"Calling RES-4 'mechanical' understates it: interval arithmetic over the full chain (J integrals, K budgets, packing counts) is a real implementation effort."* VALID-with-mitigation: relabeled honestly as mechanical-but-nontrivial; the chain's ingredients are smooth in I with certified endpoint values and monotone structure, so the sweep is an engineering task with no new mathematics — but it IS a task, registered as a candidate follow-up script with its own self-test asserts (not executable this session; sandbox shell unavailable).

(γ) *"The inventory might lull the programme: four of six 'covered' sounds like near-closure, a forbidden framing."* DISMISSED with the note's own structure: every 'covered' is qualified CONDITIONALLY with its exact scope (amended class, second cumulant, three anchors), the two open frontiers are labelled research-grade (RES-3) and deepest-analytic (RES-5), and no tier action or near-closure phrase appears anywhere. The inventory's purpose is attack-surface enumeration for T7, not status inflation.

6. Result footer

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Result ID:              B5-BEYOND-LAYER-BOUND / hlayer-residual-
                        inventory (unit U3)
Precise statement:      INVENTORY (no new theorem): the distance
                        between the amended-class conditional theorem
                        and unconditional whole-Reading-H decomposes
                        into RES-1 (H-diag, merged with RES-5 beyond
                        2nd order), RES-2 (sigma channel, covered at
                        2nd cumulant), RES-3 (unrestricted class =
                        DR-2, research-grade), RES-4 (intensity
                        interval sweep, mechanical-but-nontrivial),
                        RES-5 (matched-order to exact = GAP-2,
                        deepest), RES-6 (sea completeness, reduces to
                        RES-3). Independent open axes: {RES-3, RES-4,
                        RES-5}.
Scope:                  Survey of certified material; honest per-item
                        tiers; no quantitative claim beyond already-
                        certified constants.
Dependencies:           B5 chain (main + AddA-AddE), Math427-Math429
                        migrated records, unit U1 entry package.
Evidence grade:         SURVEY of ANALYTIC+EXECUTED material; no new
                        numbers.
Reproduction command:   python codes/vacuum/beyond_layer_gershgorin_bound.py
Expected output:        claims 192/192 PASS (existing artefact; cited
                        constants only).
Falsification gate:     Discovery of a residual axis NOT in the
                        six-item union (would void the inventory's
                        completeness claim); demonstration that a
                        listed 'covered' item is not in fact covered
                        at its stated scope.
Tier before / after:    No tier action. B5 stays T5-CANDIDATE; B1/B2
                        unchanged.
No-overclaim statement: 'Covered' always means CONDITIONALLY at the
                        stated scope. Nothing here closes RES-3,
                        RES-4, or RES-5. PDF build + linter execution
                        PENDING operator-side (sandbox shell
```

	unavailable to this agent session).
Next required action:	Unit U4 (SC-SCOPE third-cumulant lift assessment = the RES-5 quantification); RES-4 interval-sweep script as a registered candidate follow-up; DR-2 remains the RES-3 route (research-grade, off critical path).